The right ward. The last four weeks. An honest number.

A cumulative antibiogram that refreshes with the laboratory feed, shows a percentage only when the sample is honest, notices clusters and rises by itself, and answers plain-language questions by selecting from the statistic.

The task: JunctionX Turku 2026 - TYKS Laboratories

Turn laboratory susceptibility data into a continuously updated, unit-level antibiogram with anomaly detection, usable at the moment an empirical antibiotic is chosen, without touching patient-level data in the browser.

Our answer: Antibiogram Varha

One page that opens on the patient's care unit for the ward physician, infection control and stewardship, with an Ask box whose every answer is an existing cell or view. Rules the laboratory owns, in one file.

Scope honestly

A percentage only from 30 isolates; otherwise widen the period, then climb a level, and say so on the cell.

Detect

Rises and clusters against each ward's own baseline, graded, explained in one sentence.

Ask

A decision model that picks from the statistic's vocabulary and never computes a number.

Stay current

The same computation on every laboratory feed batch; the annual report comes out of the same run.

Prototype with synthetic data: no real patient appears anywhere. Team Clean Code, Dirty Wibes: Charlotte Levula, Joni Obradovic. JunctionX Turku 2026 - TYKS Laboratories, 16 to 18 October 2026.

When an antibiotic has to be chosen now, last year's hospital-wide report is the wrong answer.

Our working problem framing: three questions the statistic should answer, for three readers.

01

What is likely to work here?

The ward physician: this ward, this month, and only when there are enough isolates to say.

02

Has something changed?

Infection control: is a cluster forming right now, on which ward, and how far from the ward's usual count.

03

Where is resistance drifting?

Stewardship: year against year, county, ward group or ward, with the sample size beside every figure.

Today all three read a PDF published once a year for the whole hospital, or ask the microbiologist. The physician choosing an antibiotic needs their own ward and the last few weeks.

The framing is built from the brief and public documents (the annual Tyks resistance report, Finres), not from clinician interviews; it is a design hypothesis.

Most problems in healthcare software are not hard. They are confusing.

Our working philosophy: systems that are hard to navigate, hide the relevant number, and automate the wrong thing. The fix is design, not more features.

> ”Good design is actually a lot harder to notice than poor design, in part because good designs fit our needs so well that the design is invisible, serving us without drawing attention to itself. Bad design, on the other hand, screams out its inadequacies, making itself very noticeable.”
>
> DONALD NORMAN, THE DESIGN OF EVERYDAY THINGS

> ”There is no ’human error’, only bad design.”
>
> DONALD NORMAN

Show the relevant thing first, without a query

The page opens on the patient's ward, the specimen and the last four weeks. The annual PDF asks the physician to find the right table; this page has already found it. Good design here is a screen nobody notices.

Make the system's state legible

A percentage from 7 isolates prints exactly like one from 300; misreading it is not the reader's error. So every cell carries its n and interval, and a widened or lifted cell says what was done and why.

Automate the right thing

Recomputation, baselines and cluster detection are automated; the clinical decision is not. The Ask box selects and never answers a patient question, so it cannot be used wrongly by accident.

The same stance runs through the rest of the deck: the sufficient-sample rule, the one-sentence evidence, the launch context and the guard are design decisions before they are statistics or models.

Scope: four roles, five views, one statistic.

One shared pattern: the launch context opens the right view, every number carries its sample size, and the Ask box only selects.

ROLE	WHAT THE DEMO DOES	PURPOSE
Ward physician	Opens on the patient's care unit; organism × antibiotic matrix with % R, n and 95 % interval; the last 4 weeks beside the year; alerts on the row	Choose empirical therapy in seconds, not after a search.
Infection control	Graded alerts with timelines, the cluster table, always-alert isolates, the data-quality panel	See a cluster the day it forms.
Stewardship	Year against year per county, ward group or ward; the largest changes with their sample sizes	Steer prescribing where resistance drifts.
Lab IT	Pipeline, every rule and threshold, provenance, data quality, the model evaluation with its calibration	Own the rules, audit the numbers.

Explore, in every role

Every cell, every alert episode and every scope decision, filterable. The page holds nothing a role cannot reach.

Ask, in every role

Typed in Finnish or English, with typos; the answer is a view and a highlighted cell, never free text.

Built on public information: the laboratory system stores the requesting unit; the Varha data lake already receives laboratory results; that the patient record system can open the page with the role and care unit is assumed.

Three design decisions, each with a visible boundary.

Every obvious fix fails on the same point: one ward produces few isolates in a few weeks.

APPROACH	OUR DESIGN DECISION	WHY THIS HELPS
Scope honestly	A percentage only from 30 isolates after the one-isolate-per-patient rule. Below that the period widens (8, 13, 26, 52 weeks), then the ward group or the county is used, and the cell says what was done.	Over four weeks only 4 of 112 ward cells reach 30 isolates. A percentage from 7 isolates prints exactly like one from 300; a badge and the interval make the difference visible.
Detect against the ward's own baseline	The last 4 weeks against the preceding year on the same ward; clusters against the ward's usual count with a Poisson tail; two grades, watch and alert.	A fixed rule of “three cases in 14 days” would have alerted 27 times in three years for ESBL E. coli alone, mostly on wards where three is normal.
Ask by selecting	Rules read the request first; a fine-tuned decision model answers nine fixed questions whose options are the statistic's own vocabulary. Either side can redirect a patient-specific question.	No option for a number, a name or a sentence exists, so none can come out. A free-text model would answer the patient question too and invent a figure.

Every threshold the laboratory owns (minimum sample, window ladder, suppression, intrinsic resistance pairs, detector thresholds, always-alert list) lives in one configuration file. No model computes a number.

Show a percentage only when it is honest; otherwise show what was done.

Proposed principle: the estimator widens the period step by step, then climbs a level, and the cell always carries the reason.

PERIOD	WARD × SPECIMEN × ORGANISM	AT LEAST 30 ISOLATES	AT LEAST 10 ISOLATES
4 weeks	112	4	16
8 weeks	134	10	34
13 weeks	150	14	48
26 weeks	166	36	80
52 weeks	178	53	109

8 097 → 6 965

isolates in one year before and after “one isolate per patient”. Repeat samples are removed before any percentage is computed; every percentage then carries its n and 95 % interval.

1 · Own ward, 4 weeks
“own unit, 4 wk, n = 33”

2 · Widen: 8, 13, 26, 52 weeks
“widened to 26 weeks, n = 35”

3 · Higher level: group, county
“Inpatient wards, 26 wk, n = 54”

4 · Under 5 isolates: not shown
“not shown: too few tested isolates”

Urology urine samples hold nine organisms: one shows at its own 4-week figure, three from a widened period, five as the shared figure of the inpatient wards. 810 such decisions across the dataset, each recorded on the cell. Threshold 30 isolates (CLSI M39), synthetic data, 156 weeks.

Urology, urine, E. coli: the view a physician reads in seconds.

The example connects the honest sample, the printed interval and the comparison with the year.

SHOWN
n = 33
own unit · last 4 weeks

vs

COMPARED WITH
n = 302
same unit · calendar year 2025

AGENT	% R, 4 WK	95 % INTERVAL	YEAR 2025
nitrofurantoin	**0.0**	0.0 – 10.4	0.3
mecillinam	**3.0**	0.5 – 15.3	3.0
ceftriaxone	**9.1**	3.1 – 23.6	11.3
ciprofloxacin	**15.2**	6.7 – 30.9	14.6
trimethoprim	**21.2**	10.7 – 37.8	27.5

Wide intervals are printed, not hidden

A 33-isolate cell has wide intervals; that is why they stand beside every percentage. Urinary E. coli also by sex; everything also split into hospital- and community-onset.

Already in the right place

The page opens from the patient record interface with the care unit and specimen preselected; the ward can be changed in the sidebar. Prints on A4.

Numbers from the demo on synthetic data. Percent R is the headline everywhere, as Tyks and Finres print it; percent S is a derived tooltip.

Two planted events, both found, each explained in one sentence.

The same sentence the page shows is the evidence: counts, baseline and period, never a bare p-value.

Urology, June 2025 · rise

46.7 % R

E. coli, urine, ceftriaxone: 21 of 45 isolates in 9 June to 6 July, against 5.9 % in the preceding year (304 isolates). Four other agents alert at the same time.

Urology, June 2025 · cluster

11 patients, 0.6 expected

ESBL E. coli in 14 days where the ward's usual count is 0.6. The same outbreak seen in two independent ways.

Intensive care, March 2026 · cluster

6 patients, 0.06 expected

MRSA. The ICU has too few S. aureus isolates for any percentage, so only a count can see this; that is why there are two detectors.

DETECTOR	HOW IT WORKS	FINDINGS IN 3 YEARS
Rise in resistance	Last 4 weeks' % R against the preceding year on the same ward; watch at 10 isolates, alert at 30 isolates and a rise of at least 10 points	51 watch · **6 alerts**
Cluster	Same organism and phenotype on one ward within 14 days, against that ward's usual count; watch at 2 patients, alert at 3	9 watch · **4 alerts**
Always alert	A single CPE, VRE or meropenem-resistant Enterobacterales isolate	17

Three more episodes reach alert grade: an ampicillin rise in one health centre (2024), a 3-patient ESBL cluster and a 4-patient carbapenem-resistant Pseudomonas cluster, each explained the same way. A pilot tunes the thresholds on real history.

Rolling by construction: the demo has already run it 78 times.

The three-year scan behind the detector timelines is what production runs on every laboratory feed batch.

1 · A batch of results arrives
The results feed into the data lake exists today; target cadence every 15 to 60 minutes. In the demo, a CSV snapshot.

→

2 · Everything is rebuilt
One isolate per patient, intrinsic resistance out, every window, level and scope decision. Three years rebuild in under 30 seconds; the same input gives the same page.

→

3 · The detectors run on the window ending now
Last 4 weeks against the preceding year on the same ward, the 14-day cluster window, the always-alert list, the quality checks. The latest step is the live one.

→

4 · The page refreshes
Two aggregate JSON files are replaced; the next page load shows them. The annual report and the Finres submission come from the same run.

A rule change is replayed over three years

Every threshold lives in one file; changing one re-runs the whole history, so the microbiologist sees how many alerts a new rule would have raised before it goes live.

The feed watches itself

Isolate volume under half the ward's baseline, susceptibility results missing on over 5 % of isolates, or two EUCAST versions in one window are flagged. A broken feed shows as a quality flag, never as a silent zero.

Micro-batch is real time for this question

A rise is measured over 4 weeks and a cluster over 14 days; a result processed within the hour changes nothing a result processed within the second would. Batches keep patient-level data out of the browser.

Demo: the scan steps every 14 days so the timelines carry three years of history (78 steps, 104 graded findings, 797 quality rows, 6 flagged). Production: every batch is a step, and the late-results check this dataset cannot carry is named rather than omitted.

Ask in your own words; the answer is always a place in the statistic.

Rules read the request first; a fine-tuned decision model answers nine fixed questions in one pass and never writes text.

1 · The request

“ecoli virtsa uro naiset 4vk”

any wording, Finnish or English, typos included

→

2 · Laya reads it once

An open 322-million-parameter encoder, fine-tuned on about 4 500 labelled requests, scores nine questions in one pass: 0.6 s on a laptop CPU. Each answer is a probability over a closed list.

→

type aggregate	**view** clinician	**organism** E. coli
specimen urine	**ward** URO	**sex** women
onset none	**period** 4 weeks	**antibiotic** none

The guard

“My patient has E. coli in urine, what should I start?” gets no answer: the question is redirected to the physician's judgement. The rules run first; if they or the model say patient-specific, or the model's probability of it exceeds 0.2, nothing is answered.

Why it cannot invent a number

The options are the statistic's own vocabulary: nine organisms, three specimen types, the wards, the periods, thirteen antibiotics. There is no option for a number, a name or a sentence. The page then shows the selected cell with its n and interval.

What the page does with it

Opens the right view on the right ward, highlights the organism and the antibiotic, collapses the matrix to the asked rows, and shows the interpretation beside the prompt with every field and its source.

A patient-specific question is redirected and never answered. The model is open (Laya, Apache 2.0), fine-tuned by us, and runs on a laptop CPU; a larger local language model was measured and not shipped.

Rules are exact but rigid; the fine-tuned model reads everyday phrasing; together, both.

Model tests on 170 labelled questions (Finnish and English), counts first, percentages after.

INTERPRETER	QUESTION TYPE CORRECT	ALL SEARCH TERMS CORRECT	PATIENT-SPECIFIC REDIRECTED	PER QUESTION
Rules in the browser	170 / 170	170 / 170	36 / 36	under 1 ms
Laya before fine-tuning	31 / 170 · 18 %	36 / 170 · 21 %	9 / 36	0.65 s CPU
Laya, first fine-tune (367 hand-written examples)	163 / 170 · 95.9 %	157 / 170 · 92.4 %	33 / 36	0.62 s CPU
Laya, final (4 500 examples, 23 min, one GPU)	169 / 170 · 99.4 %	168 / 170 · 98.8 %	35 / 36	0.64 s CPU
Qwen3-30B-A3B instruct, local LLM, 1 GPU	168 / 170 · 98.8 %	142 / 170 · 83.5 %	36 / 36	1.3 s GPU
Qwen3.6-27B, local LLM, 2 GPUs	168 / 170 · 98.8 %	164 / 170 · 96.5 %	36 / 36	6.0 s GPU
Rules + Laya: the shipped guard	**170 / 170 · 100 %**	**169 / 170 · 99.4 %**	**36 / 36**	**0.64 s CPU**

Never seen in training

60 stress questions written after training: type 57 / 60, all terms 49 / 60, redirected 10 / 12, false redirects 0 / 48. Held-out 41 of the 170: type 40 / 41, terms 39 / 41, calibration error 0.024.

Why rules come first

Either side can redirect; a keyword found by the rules is never overridden. The two patient questions the model misses ask for “all patients' records”; the rules call them off-topic, so nothing is answered.

Light to run

322 million parameters, about 650 MB, 0.6 s per question on a laptop CPU; fine-tuning took 23 minutes on one GPU. The Qwen models need one or two GPUs and are not shipped.

We wrote the 170 and the 60 test questions ourselves, so the figures describe intended behaviour, not independent accuracy.

Explore the interactive demo ↗ · 11 / 15

What a yearly report or a dashboard cannot do, and this page does.

Four properties, each visible on the page rather than claimed.

FOR THE READER

Opens in the right place

- Launch context picks the view, ward and specimen; nothing to type
- One matrix for the physician, alerts for infection control, trends for stewardship
- Finnish and English, phone width, A4 print
- Everyday phrasing, typos included, opens the right cell

FOR THE NUMBER

Says how sure it is

- One isolate per patient, intrinsic resistance out, 30 isolates before a percentage
- n and the 95 % interval beside every figure
- A widened or lifted cell says what was done: 810 recorded decisions
- Under 5 isolates: not shown, and the cell says why

FOR THE CHANGE

Watches by itself

- Rise and cluster detectors against each ward's own baseline
- Single CPE, VRE or carbapenem-resistant isolates always alert
- Graded watch and alert, one explaining sentence in both languages
- Both planted events found; a handful of findings elsewhere

FOR THE LABORATORY

Stays current, stays bounded

- The same computation on every feed batch; the annual report from the same run
- Every rule in one file the laboratory owns and can replay
- No patient-level field reaches the browser
- Ask selects and never computes; the patient question is refused

A dashboard shows whatever is asked of it. This page decides what is honest to show, notices what has changed, and knows which question it must not answer.

Each property is checked in the demo: 68 example questions open the intended view, 810 scope decisions are printed on the cells, 104 graded findings are explained, both planted events are found. The effect on prescribing is for a pilot to measure.

The demo establishes the mechanics; a pilot must establish the effect.

What is shown to work on synthetic data, and what only real data on a real ward can answer.

What the demo establishes

- Both planted events are found at alert grade and explained in one sentence
- 810 scope decisions are visible on the cells; no percentage without n and interval
- 68 example questions, with typos and mixed language, open the intended view; every patient-specific one is refused
- Every number traces to the exported JSON; no patient-level field reaches the browser

What a pilot must establish

- How often results reach the lake, and whether S/I/R is there at isolate level
- A maintained mapping from laboratory unit codes to care units
- Detector thresholds tuned on real history, and the real false-alert rate
- Whether physicians change an empirical choice, and whether infection control acts earlier

1 · Now
Prototype on synthetic data; all rules in one file for the microbiologist to approve

→

2 · Pilot
Real data in the lake; thresholds tuned on real history; the ward mapping table

→

3 · Deployment
Opened from the patient record system; the Finres submission from the same computation

→

4 · Separate project
The patient-specific module, as a medical device; not part of this product

A surveillance tool on the same footing as the annual report: the laboratory monitoring its own operations, output aggregated. A patient-specific recommendation would be a medical device and is not built. Effects on prescribing and on outbreak response are unmeasured.

Evidence and a direct route to the demo.

Open the page, type a question, inspect every number's sample size, and read the rules the laboratory would own.

OPEN THE DEMO

https://joniobradovic.github.io/junctionx_tykslab_submission/?role=clinician&unit=URO&specimen=urine

Opens in the ward physician's view on urology urine samples; add lang=fi for Finnish, or drop the parameters for the role selector. The Ask box runs the rules in the browser; the decision model runs locally from the repository.

CODE, DATA AND MODEL

https://github.com/JoniObradovic/junctionx_tykslab_submission

Key primary sources

- Tyks, “Antibioottiresistenssitilanne Varsinais-Suomessa 2024” (tyks.fi)
- THL Finres national surveillance
- CLSI M39 cumulative antibiogram guideline
- EUCAST expected resistant phenotypes
- Act on the Secondary Use of Health and Social Data 552/2019, section 41

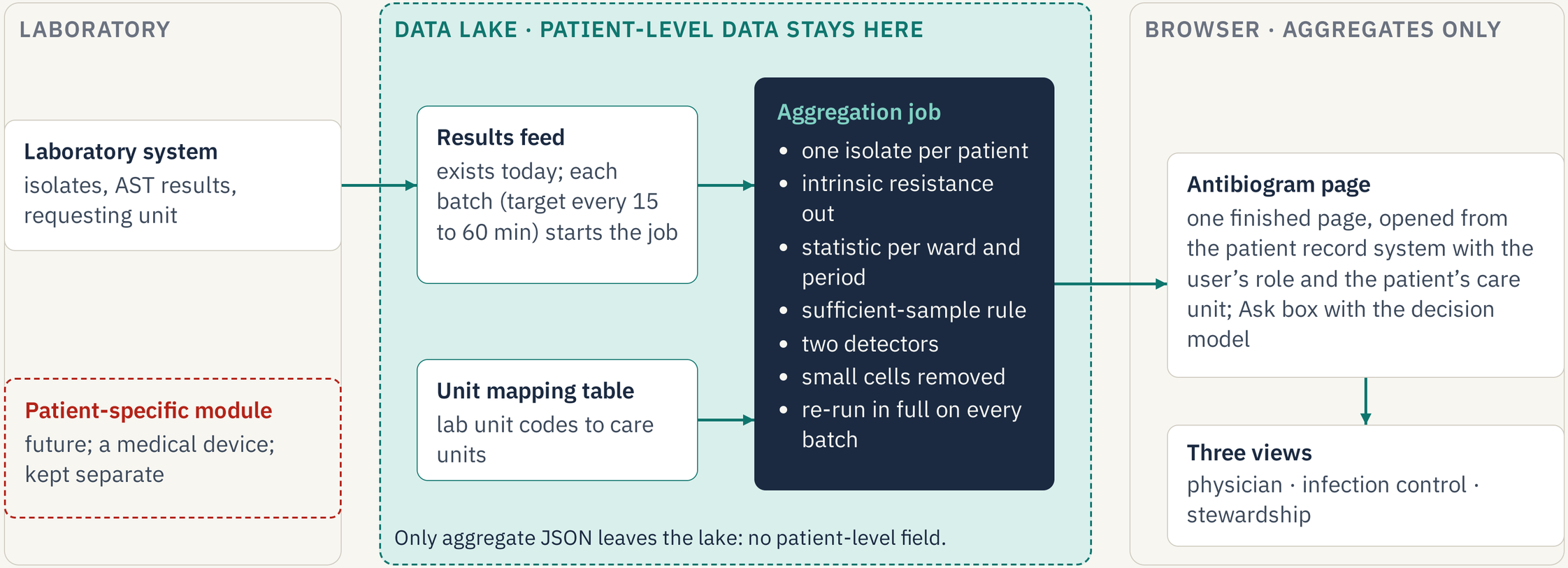
Keep assumptions visible

- All data is synthetic; the two events were planted
- Lake cadence and the launch context are assumed
- Model tests are a specification by the authors, not independent accuracy

Challenge: supplied brief. Problem framing: our hypothesis. Public sources checked 17 October 2026. The code runs on a laptop: page from any static server, the decision model on CPU.

Appendix: how it would run in production.

Three zones. Patient data stays in the lake; only the finished statistic reaches the browser.



Not tied to Weblab, the Varha lake or Omni360: the design needs isolate-level results with a requesting unit, somewhere to compute, and a place to open the page from. The patient-specific module is drawn apart on purpose.